Measuring Inequality and Bias in Mortgage Pricing

CS 4828 Project Proposal

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1 Team Members

Name	Role
Jack Leddy	Model development, programming
Thomas Henning	Data preprocessing, model development
Yuzhuo Zhang	Feature and residual and fairness analysis

2 Description of the Problem

Mortgage interest rates depend on factors such as income, debt-to-income ratio, loan-to-value ratio, loan amount, loan type, and market conditions. However, previous research has shown that borrowers with similar financial risk can still receive different mortgage prices across demographic groups (Bartlett et al.). This raises an important question: are similarly situated borrowers being treated equally?

The goal of this project is to use machine learning to measure potential inequality or bias in mortgage pricing. We will train a model to predict the *expected market interest rate* of a mortgage using publicly available Home Mortgage Disclosure Act (HMDA) data. Protected demographic information, such as race, ethnicity, and sex, as well as lender identity, will not be used to train the main model. Instead, the model will use financial, loan, geographic, and time-related characteristics.

For each loan, we will calculate a pricing residual:

$$e_i = r_i - \hat{r}_i$$

where r_i is the actual interest rate and $\hat{r}_i$ is the predicted rate. A positive residual means that a borrower received a higher rate than expected for a similar loan.

We will compare these residuals across demographic groups and lenders. If certain groups consistently receive higher-than-expected rates, this may suggest inequality in mortgage pricing. Lender-level analysis may also show whether certain institutions produce larger or more uneven pricing differences.

The model does not determine a perfectly "fair" interest rate because historical market data may already contain bias. Instead, it approximates expected pricing and uses deviations from that expectation to identify possible inequality. These patterns would not strictly prove discrimination, but could identify areas that deserve further investigation.

3 Preliminary Plan

1. **Data Collection and Cleaning (Weeks 1–2):** Obtain HMDA data; restrict to comparable products (first-lien, owner-occupied, 30-year fixed conventional loans); handle missing values and outliers.
2. **Feature Engineering (Weeks 2–3):** Build financial, geographic, and time features; control for macro rate movements using rate-lock quarter and the 10-year Treasury yield.
3. **Model Development (Weeks 4–6):** Train linear regression, random forest, and neural network models; select the best via cross-validated MAE/RMSE.
4. **Residual and Fairness Analysis (Weeks 7–8):** Compute residuals; compare across demographic groups and lenders using t-tests and residual regressions; verify robustness across all three models.
5. **Evaluation and Write-up (Weeks 9–10):** Interpret results, compare with prior work, discuss limitations; one week of buffer before the Nov 25 deadline.

4 References

Bartlett, Robert P., et al. "Consumer-Lending Discrimination in the FinTech Era." *Journal of Financial Economics*, vol. 143, no. 1, 2022, pp. 30–56. https://doi.org/10.1016/j.jfineco.2021.05.047.

Fuster, Andreas, et al. "Predictably Unequal? The Effects of Machine Learning on Credit Markets." *The Journal of Finance*, vol. 77, no. 1, 2022, pp. 5–47. https://doi.org/10.1111/jofi.13090.

Das, Sanjiv, Richard Stanton, and Nancy Wallace. "Algorithmic Fairness." *Annual Review of Financial Economics*, vol. 15, 2023, pp. 565–593. https://doi.org/10.1146/annurev-financial-110921-125930.